Fatima Jinnah Women University, Rawalpindi

Lab #1 : GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

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2024-BSE-10

Course Code:

SE-665

Section:

B

GitHub Username:

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Department of Software Engineering(BSE-5)

Getting Started

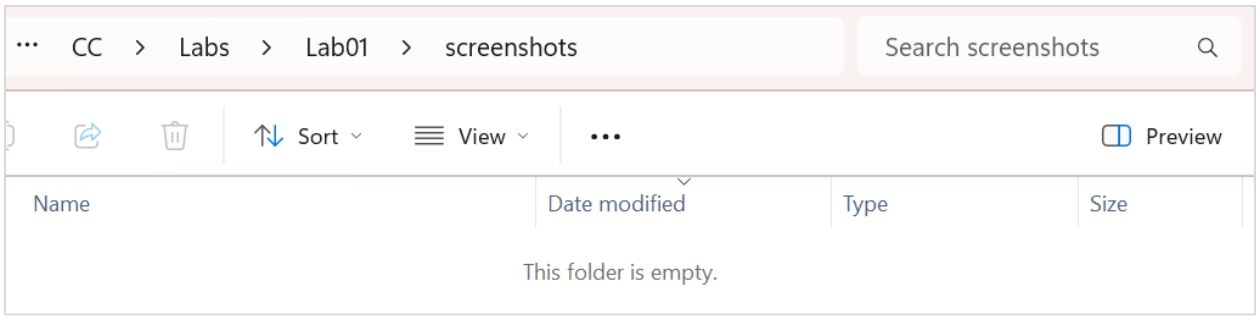


Figure 1: Local directory structure (Labs/Lab01/screenshots/) created for the Cloud Computing lab repository.

Task 1: Create and Configure Your GitHub Account

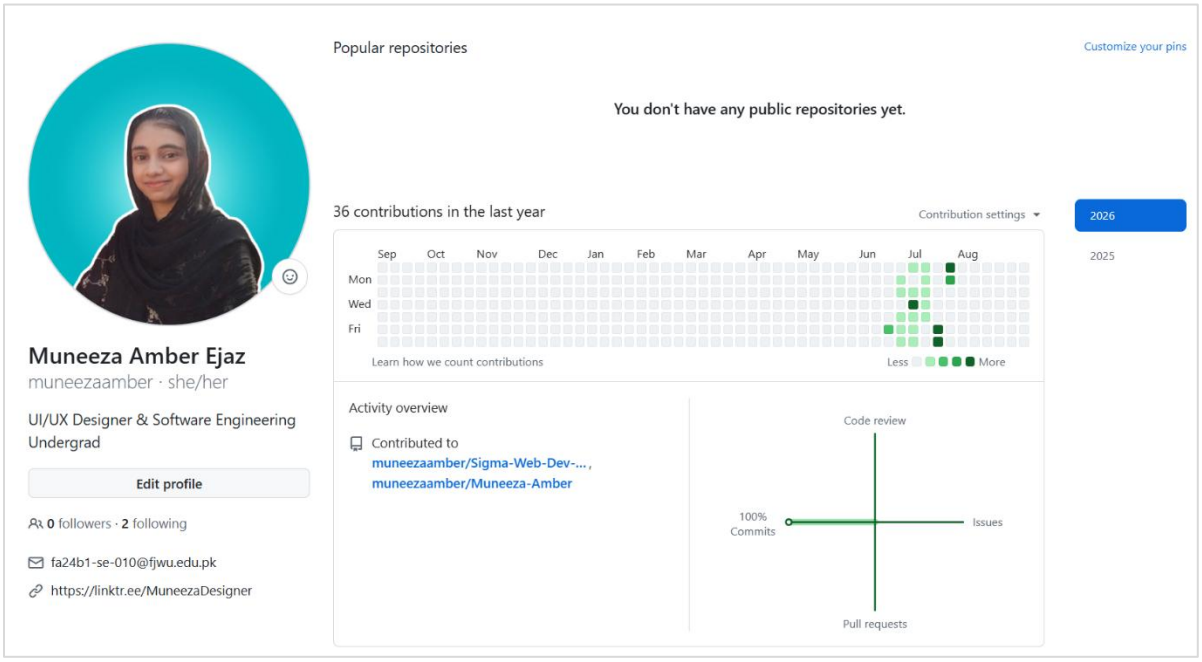


Figure 2: GitHub profile displaying the configured professional username and profile details.

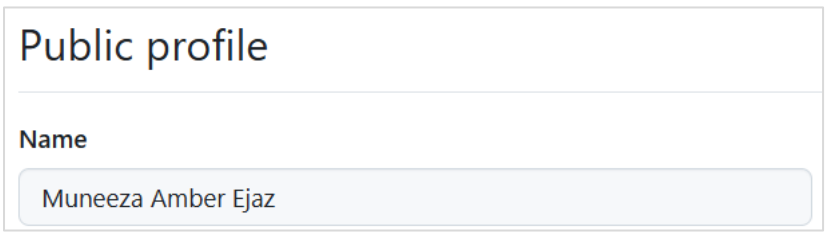


Figure 3: GitHub Public Profile settings showing the completed Name field with the actual name.

Task 2: Register in the Student Marks Portal

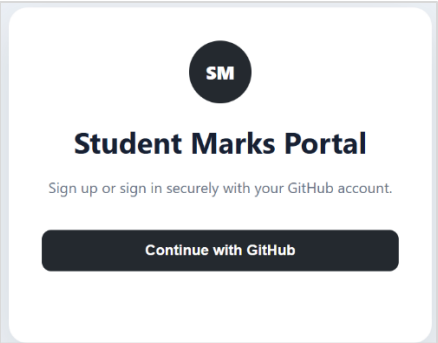


Figure 4: Student Marks Portal login page displaying the GitHub authentication option.

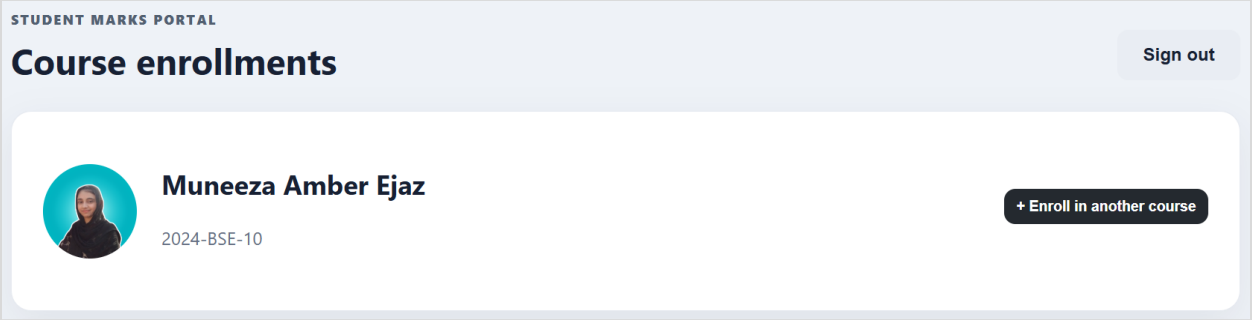


Figure 5: Course enrollments dashboard verifying the linked GitHub profile information and registration number.

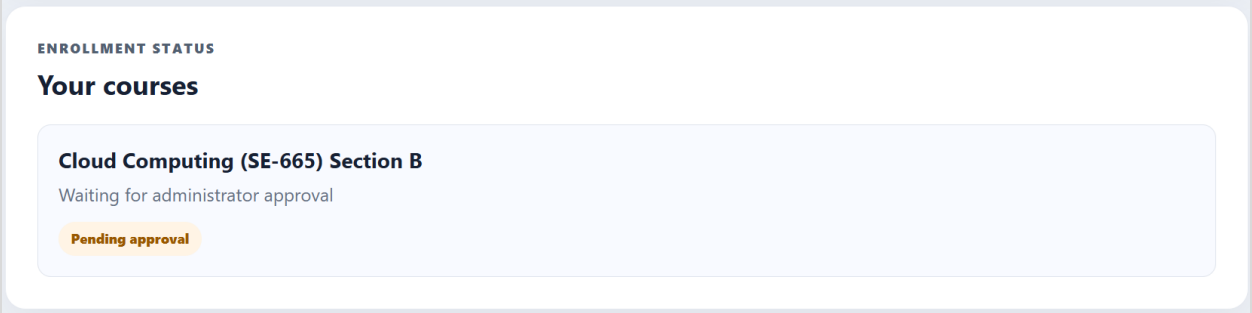


Figure 6: Enrollment status showing the submitted Cloud Computing course registration pending administrator approval.

Task 3: Download VMware Workstation Pro and Ubuntu Server

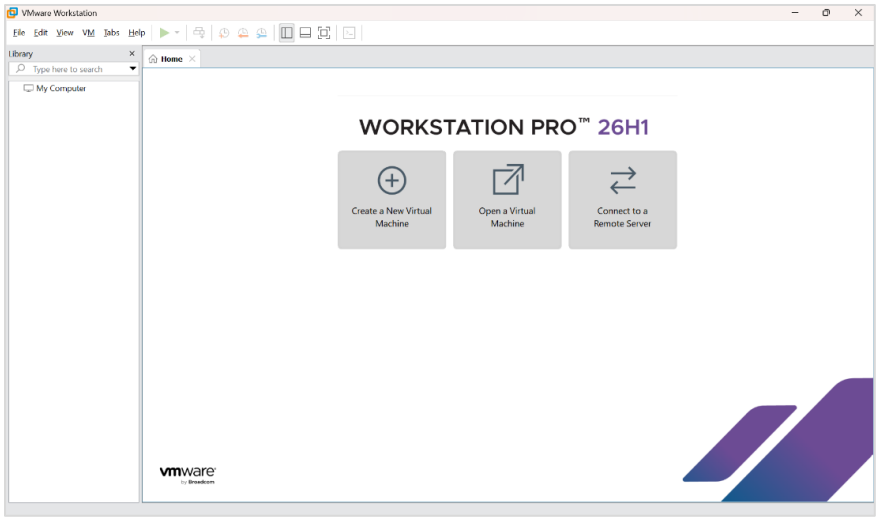


Figure 7: VMware Workstation Pro interface running successfully on the host machine.



Figure 8: The downloaded Ubuntu Server LTS ISO file ready for virtual machine installation.

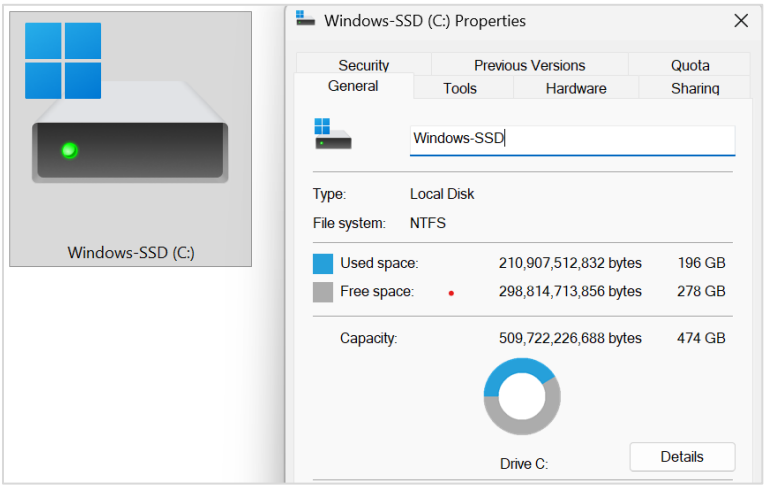


Figure 9: Local disk properties verifying sufficient available free space (over 20 GB) for the virtual machine.

Task 4: Create the Ubuntu Server Virtual Machine



Figure 10: The New Virtual Machine Wizard initiated in VMware Workstation Pro.

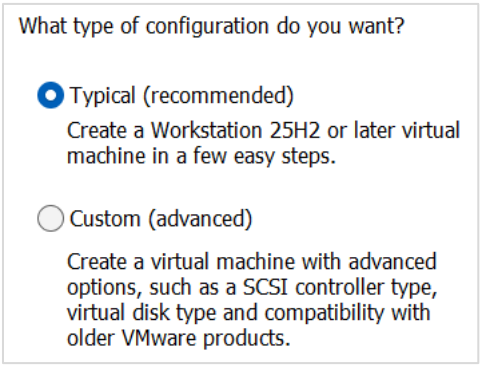


Figure 11: Selection of the Typical (recommended) configuration for the new virtual machine.

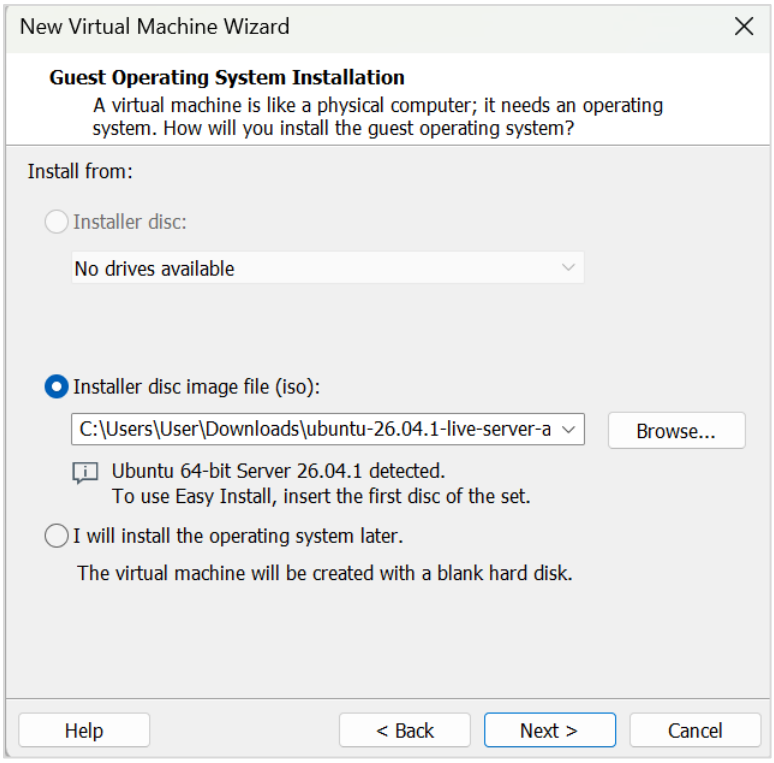


Figure 12: The downloaded Ubuntu Server ISO file selected as the installer disc image.

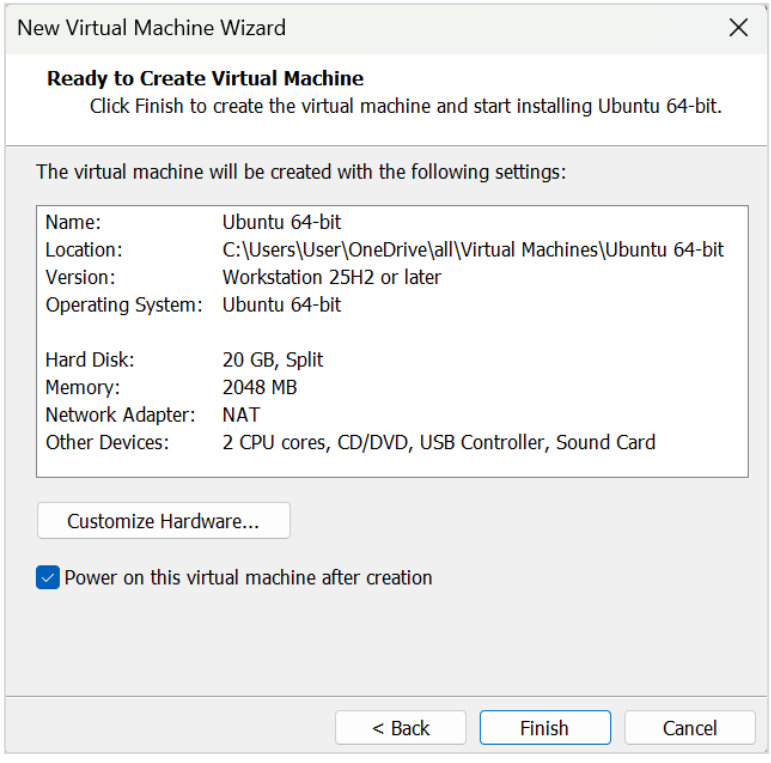


Figure 13: Final virtual machine configuration summary confirming the allocated resources.

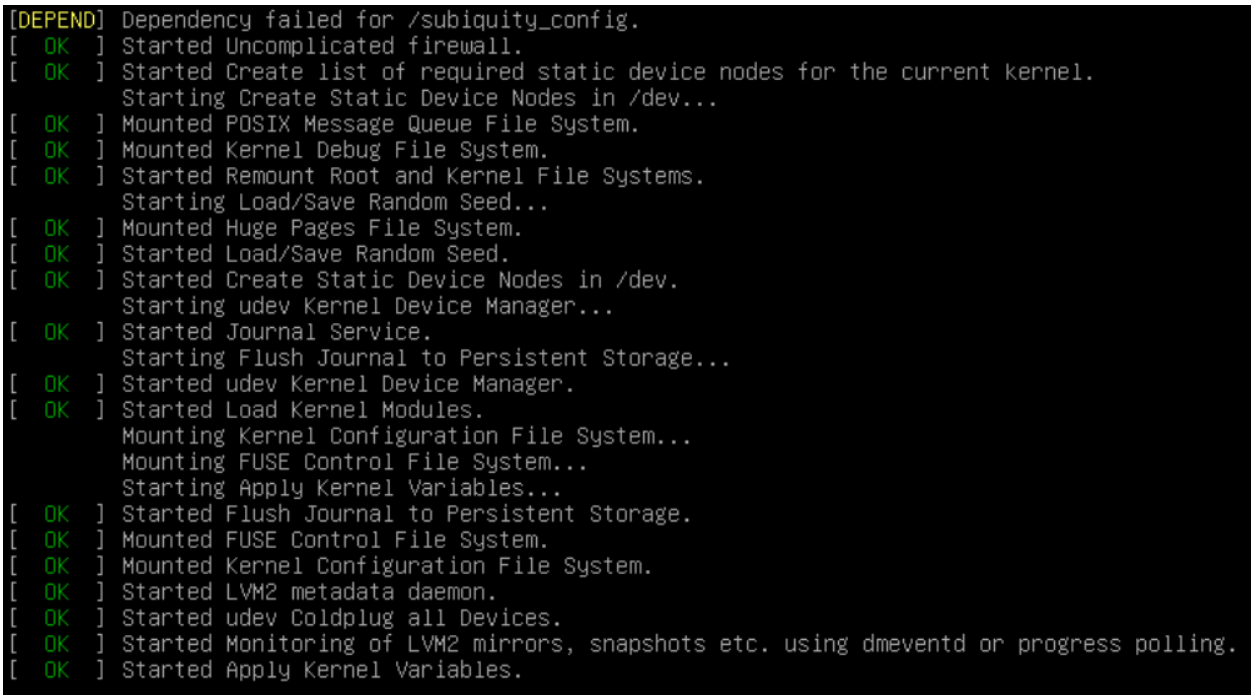


Figure 14: The initial boot sequence of the Ubuntu Server installer.

Task 5: Choose the Installation Language

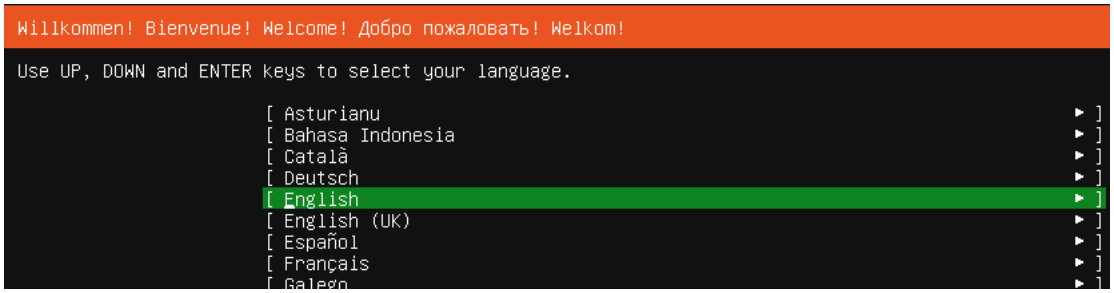


Figure 15: Selection of English as the primary installation language for the server.

Task 6: Configure the Keyboard Layout



Figure 16: Configuring the system keyboard layout to English (US).



Figure 17: Setting the appropriate keyboard variant to English (US).

Task 7: Choose the Ubuntu Server Installation Type

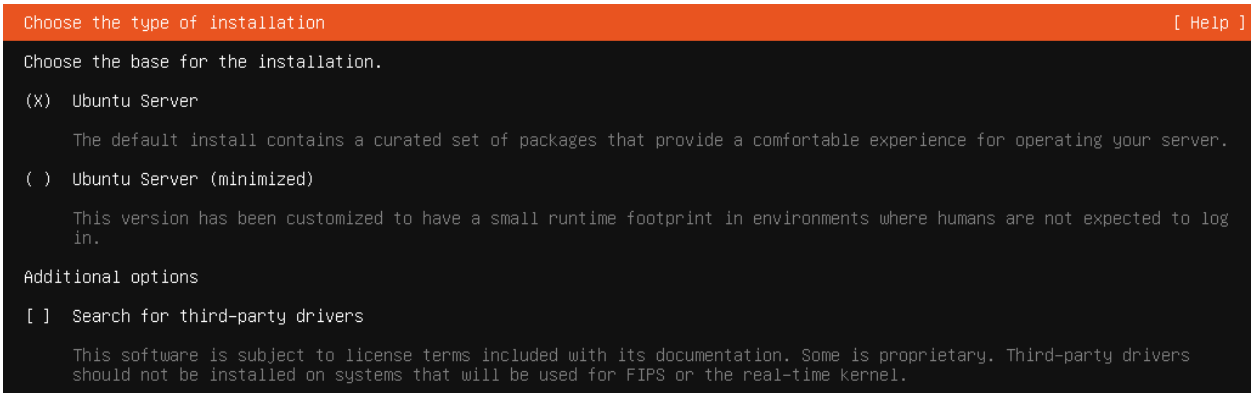


Figure 18: Choosing the standard Ubuntu Server installation base for the Cloud Computing lab.

Task 8: Configure the Network

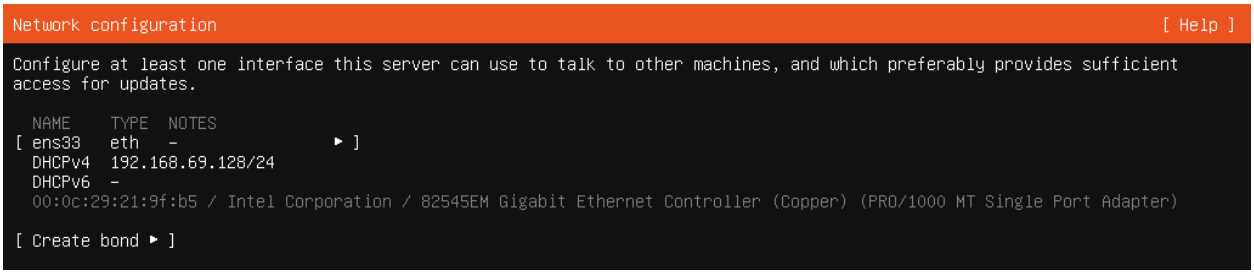


Figure 19: Network configuration screen displaying the detected ens33 ethernet interface.

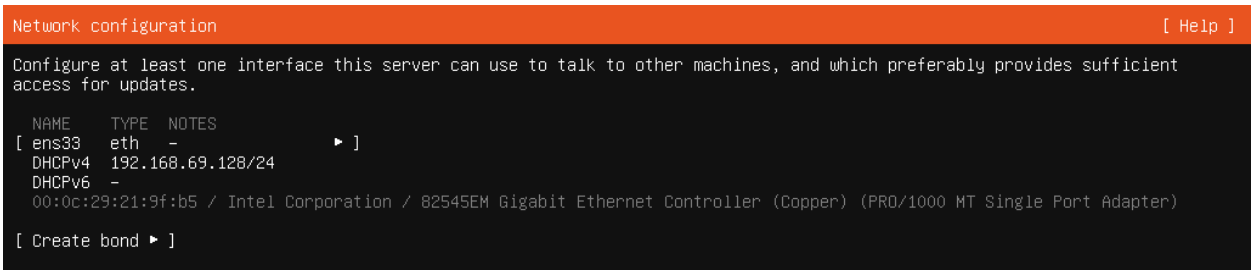


Figure 20: Confirmation of the IPv4 address assigned to the server automatically via DHCP.

Task 9: Configure Storage

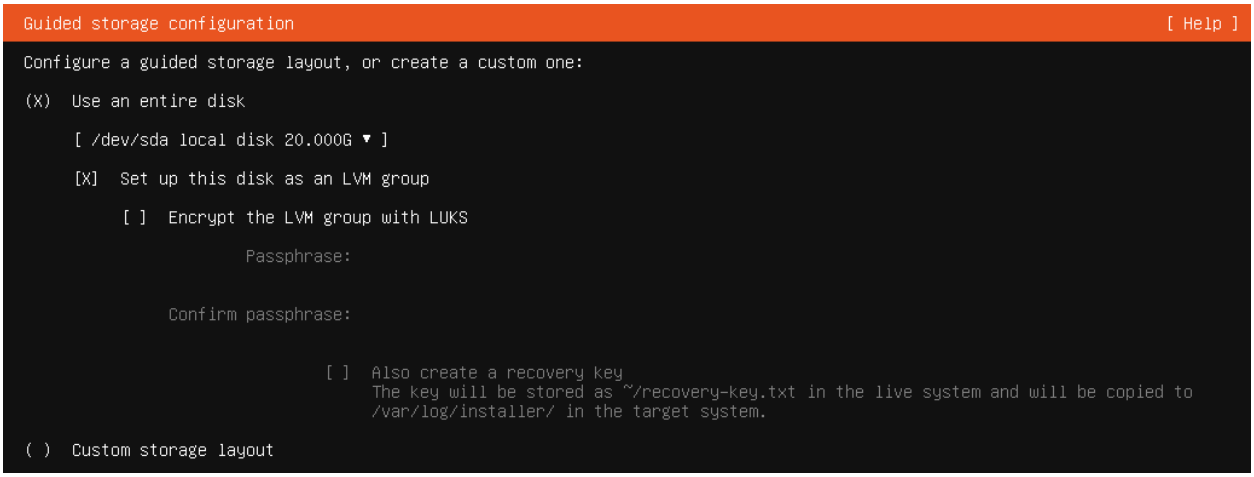


Figure 21: Selection of the guided storage option to utilize the entire VMware virtual disk.

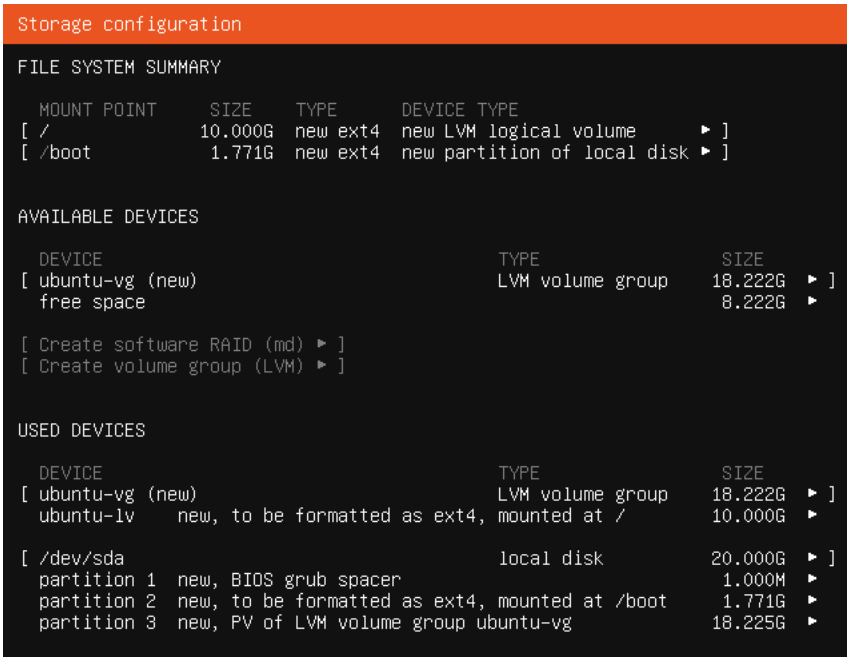


Figure 22: File system summary detailing the proposed storage configuration before formatting.

Task 10: Select the Installation Disk

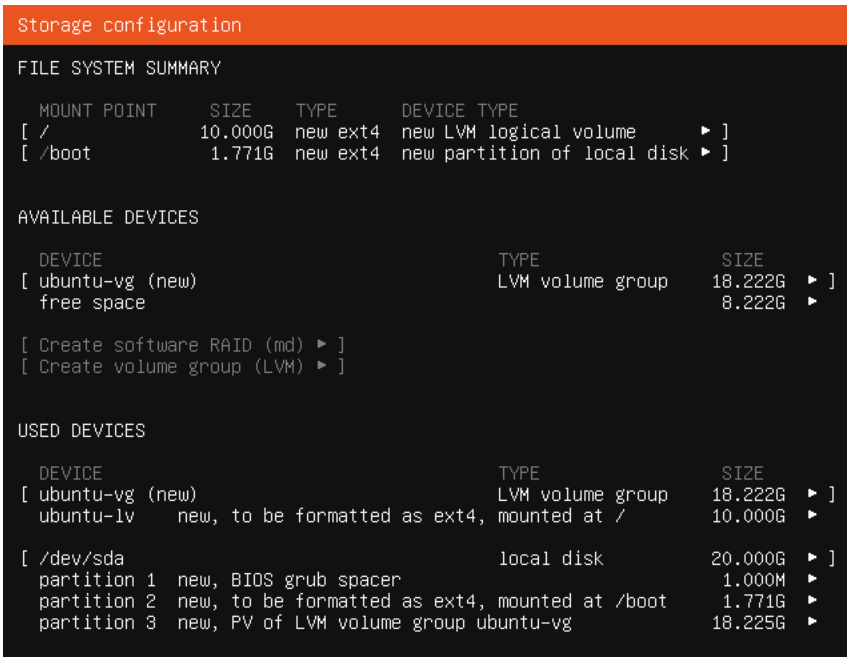


Figure 23: Selection of the 20 GB virtual disk created specifically for the Ubuntu Server installation.

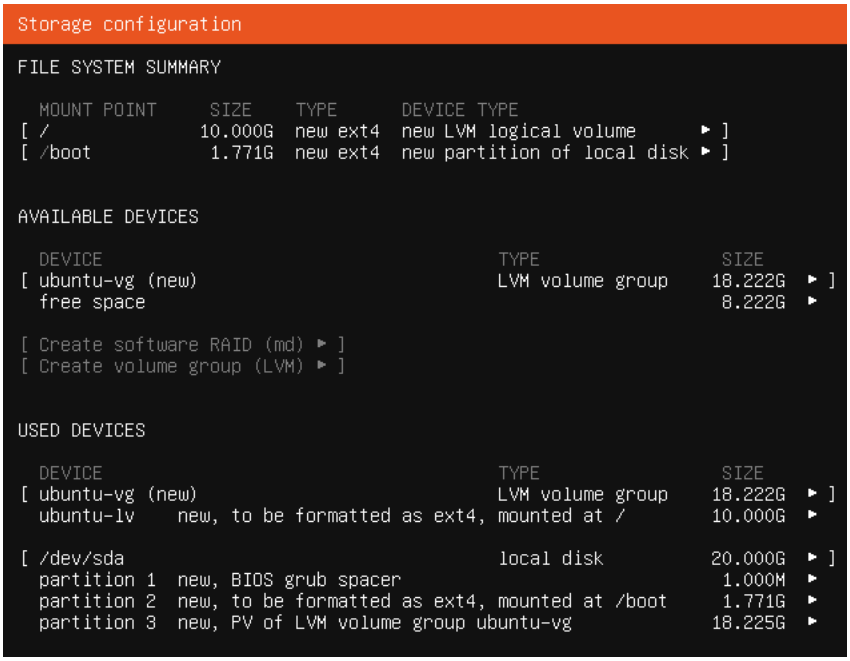


Figure 24: Verification of the displayed disk size and device type to ensure the correct VMware virtual disk is targeted.

Task 11: Review the Partitions

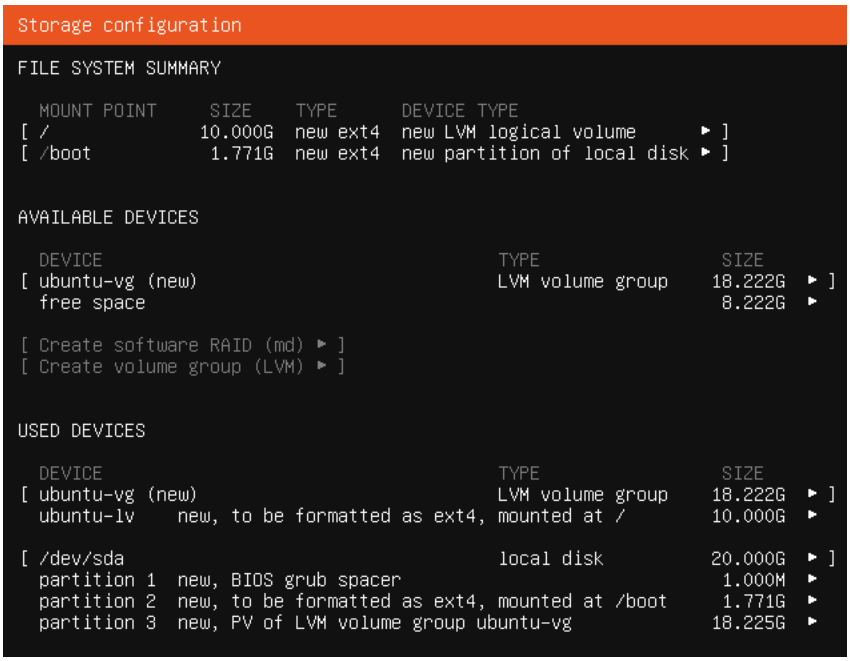


Figure 25: Review of the complete partition layout proposed by the installer to confirm it belongs to the virtual environment.

Task 12: Confirm the Storage Changes

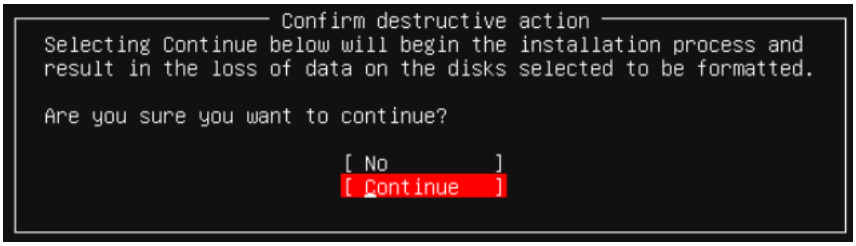


Figure 26: The destructive-action warning prompt triggered before formatting the selected virtual disk.

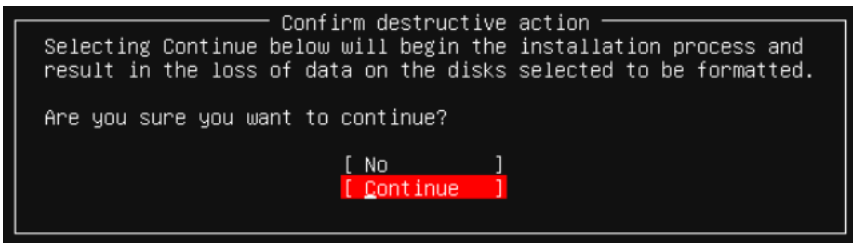


Figure 27: Confirmation to proceed with the storage changes and begin the installation process.

Task 13: Configure the Ubuntu Profile



Figure 28: Entry of the complete actual name in the Ubuntu profile configuration.



Figure 29: Configuration of the server name as 'ubuntu'.



Figure 30: Entry of the exact GitHub username to serve as the system username.

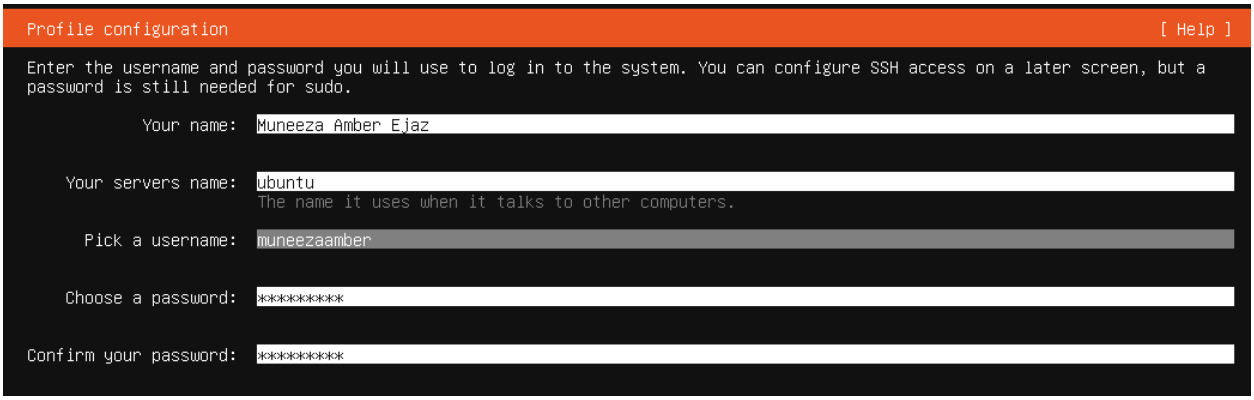


Figure 31: Review of the fully completed Ubuntu profile setup before proceeding.

Task 14: Complete the Software Installation

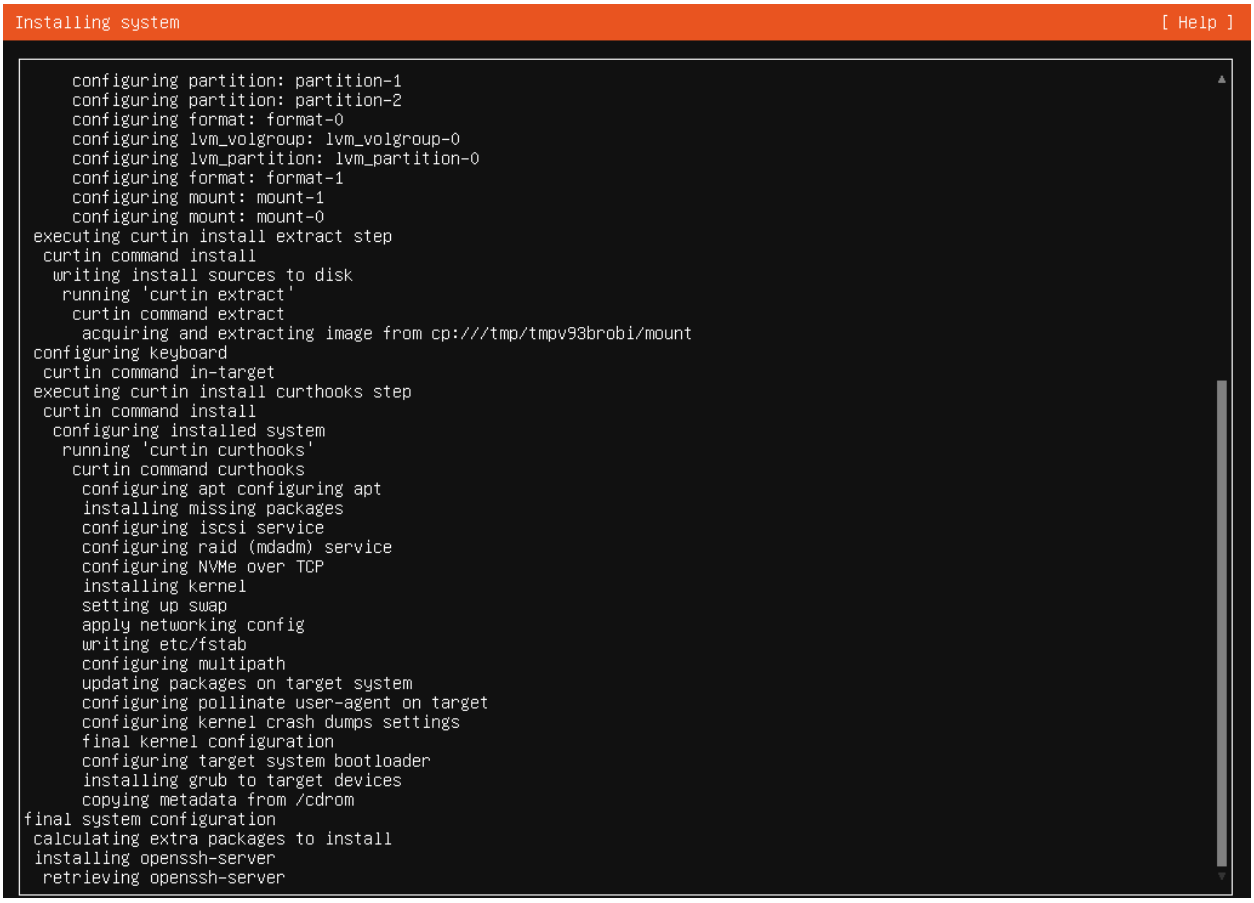


Figure 32: System installation process displaying the background package configuration.

Task 15: Reboot and Verify the Ubuntu Identity

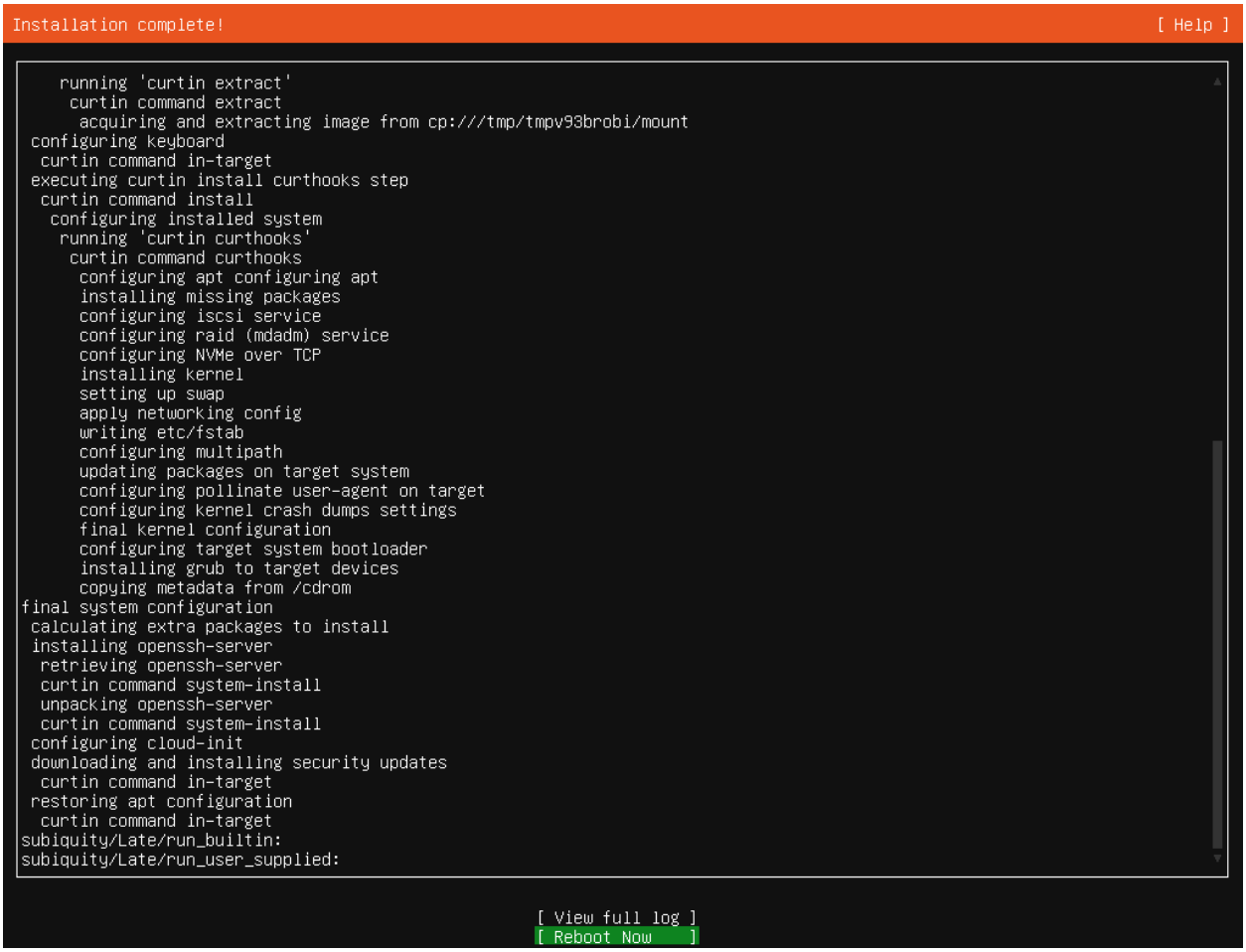


Figure 33: The installation complete screen prompting for a system reboot.



Figure 34: The initial Ubuntu Server login prompt after a successful system boot.

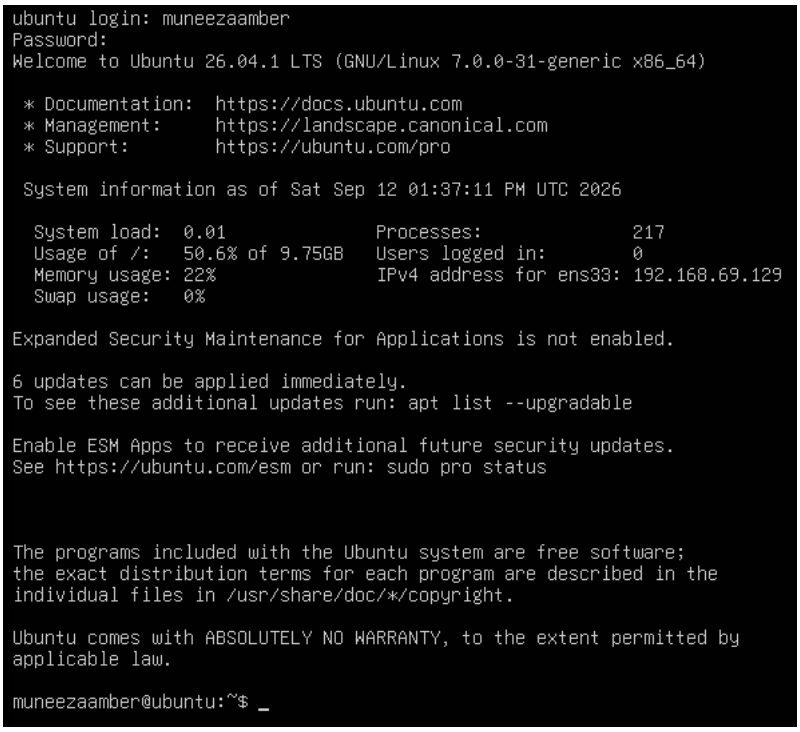


Figure 35: Successful local terminal login displaying the system welcome message.



Figure 36: Local terminal prompt verifying the required username and server identity.

Task 16: Find the Ubuntu Server IP Address

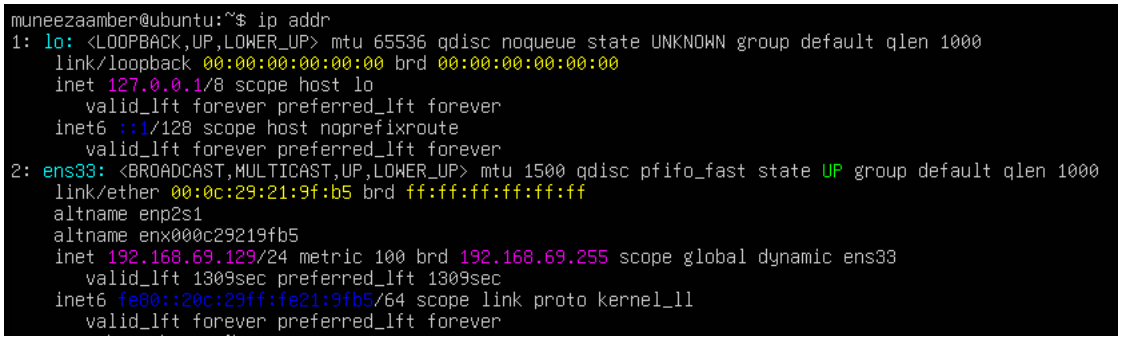


Figure 37: Execution of the ip addr command to retrieve network interface details.

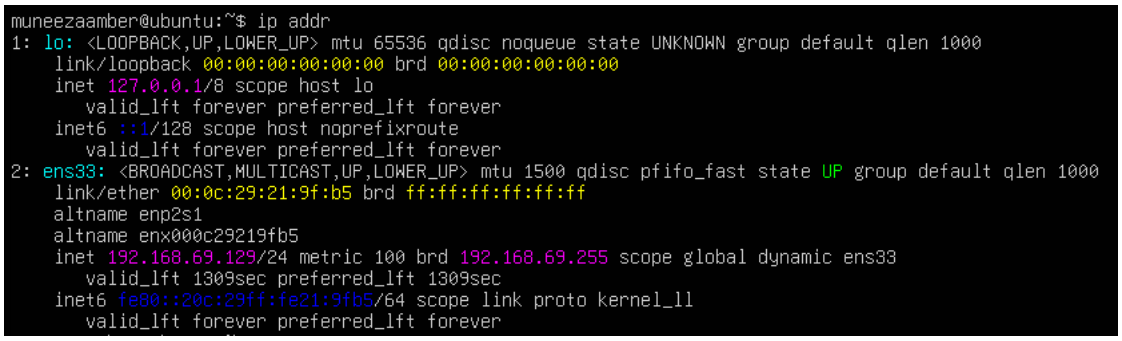


Figure 38: Identification of the assigned IPv4 address for the active network interface.

Task 17: Connect to Ubuntu Server from Windows

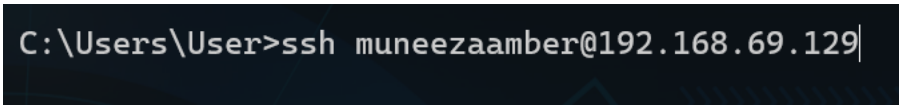


Figure 39: Execution of the remote SSH connection command from the Windows Command Prompt.



Figure 40: Confirmation and acceptance of the Ubuntu server's security fingerprint.

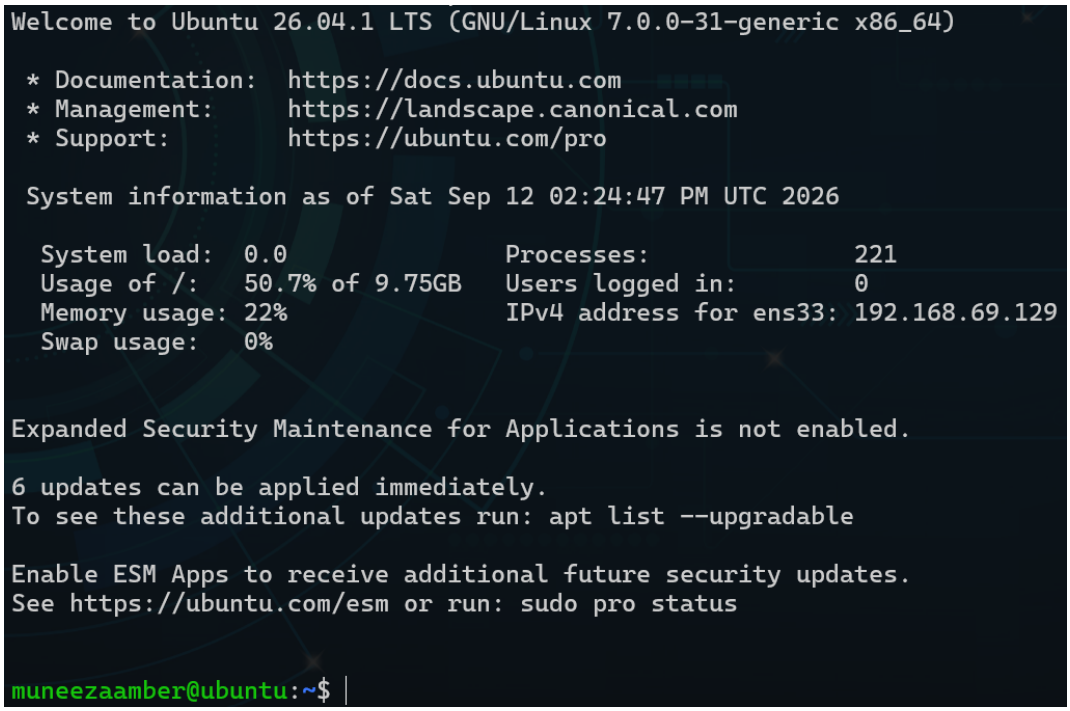


Figure 41: Successful remote SSH login displaying the Ubuntu Server welcome message.



Figure 42: Remote SSH terminal prompt verifying the user identity from the Windows host.